the right triangle there doubtless arose questions of puzzling subtlety. Thus, given a number equal to the side of an isosceles right triangle, to find the number which the hypotenuse is equal to. The side may have been taken equal to 1, 2, $\frac{3}{2}$, $\frac{6}{5}$, or any other number, yet in every instance all efforts to find a number exactly equal to the hypotenuse must have remained fruitless. The problem may have been attacked again and again, until finally "some rare genius, to whom it is granted, during some happy moments, to soar with eagle's flight above the level of human thinking," grasped the happy thought that this problem cannot be solved. In some such manner probably arose the theory of *irrational quantities*, which is attributed by Eudemus to the Pythagoreans. It was indeed a thought of extraordinary boldness, to assume that straight lines could exist, differing from one another not only in length,—that is, in quantity,—but also in a quality, which, though real, was absolutely invisible. [7] Need we wonder that the Pythagoreans saw in irrationals a deep mystery, a symbol of the unspeakable? We are told that the one who first divulged the theory of irrationals, which the Pythagoreans kept secret, perished in consequence in a shipwreck. Its discovery is ascribed to Pythagoras, but we must remember that all important Pythagorean discoveries were, according to Pythagorean custom, referred back to him. The first incommensurable ratio known seems to have been that of the side of a square to its diagonal, as $1 : \sqrt{2}$. **Theodorus of Cyrene** added to this the fact that the sides of squares represented in length by $\sqrt{3}$, $\sqrt{5}$, etc., up to $\sqrt{17}$, and Theætetus, that the sides of any square, represented by a surd, are incommensurable with the linear unit. **Euclid** (about 300 B.C.), in his *Elements*, X. 9, generalised still further: Two magnitudes whose squares are (or are not) to one another as a square number to a square number are commensurable (or incommensurable), and conversely. In the tenth book, he treats of incommensurable quantities at length. He investigates every possible variety of lines which can be represented by $\sqrt{\sqrt{a} \pm \sqrt{b}}$, a and b representing two commensurable lines, and obtains 25 species. Every individual of every species is incommensurable with all the individuals of every other species. “This book,” says De Morgan, “has a completeness which none of the others (not even the fifth) can boast of; and we could almost suspect that Euclid, having arranged his materials in his own mind, and having completely elaborated the tenth book, wrote the preceding books after it, and did not live to revise them thoroughly.” [9] The theory of incommensurables remained where Euclid left it, till the fifteenth century.

Euclid devotes the seventh, eighth, and ninth books of his *Elements* to arithmetic. Exactly how much contained in these books is Euclid’s own invention, and how much is borrowed from his predecessors, we have no means of knowing. Without doubt, much is original with Euclid. The *seventh book* begins with twenty-one definitions. All except that for ‘prime’ numbers are known to have been given by the Pythagoreans. Next follows a process for finding the G.C.D. of two or more numbers. The *eighth book* deals with numbers in continued proportion, and with the mutual relations of squares, cubes, and plane numbers. Thus, XXII., if three numbers are in continued proportion, and the first is a square, so is the third. In the *ninth book*, the same subject is continued. It contains the proposition that the number of primes is greater than any given number.

After the death of Euclid, the theory of numbers remained almost stationary for 400 years. Geometry monopolised the attention of all Greek mathematicians. Only two are known to have done work in arithmetic worthy of mention. **Eratosthenes** (275–194 B.C.) invented a 'sieve' for finding prime numbers. All composite numbers are 'sifted' out in the following manner: Write down the odd numbers from 3 up, in succession. By striking out every third number after the 3, we remove all multiples of 3. By striking out every fifth number after the 5, we remove all multiples of 5. In this way, by rejecting multiples of 7, 11, 13, etc., we have left prime numbers only. **Hypsicles** (between 200 and 100 B.C.) worked at the subjects of polygonal numbers and arithmetical progressions, which Euclid entirely neglected. In his work on 'risings of the stars,' he showed (1) that in an arithmetical series of $2n$ terms, the sum of the last n terms exceeds the sum of the first n by a multiple of n^2; (2) that in such a series of $2n + 1$ terms, the sum of the series is the number of terms multiplied by the middle term; (3) that in such a series of $2n$ terms, the sum is half the number of terms multiplied by the two middle terms. [6]

For two centuries after the time of Hypsicles, arithmetic disappears from history. It is brought to light again about 100 A.D. by **Nicomachus**, a Neo-Pythagorean, who inaugurated the final era of Greek mathematics. From now on, arithmetic was a favourite study, while geometry was neglected. Nicomachus wrote a work entitled *Introductio Arithmetica*, which was very famous in its day. The great number of commentators it has received vouch for its popularity. Boethius translated it into Latin. Lucian could pay no higher compliment to a calculator than this: "You reckon like Nicomachus of Gerasa." The *Introductio Arithmetica* was the first exhaustive work in which arithmetic was treated quite independently of geometry. Instead of drawing lines, like Euclid, he illustrates things by real numbers. To be sure, in his book the old geometrical nomenclature is retained, but the method is inductive instead of deductive. "Its sole business is classification, and all its classes are derived from, and exhibited by, actual numbers." The work contains few results that are really original. We mention one important proposition which is probably the author's own. He states that cubical numbers are always equal to the sum of successive odd numbers. Thus, $8 = 2^3 = 3+5$, $27 = 3^3 = 7+9+11$, $64 = 4^3 = 13+15+17+19$, and so on. This theorem was used later for finding the sum of the cubical numbers themselves. **Theon** of Smyrna is the author of a treatise on "the mathematical rules necessary for the study of Plato." The work is ill arranged and of little merit. Of interest is the theorem, that every square number, or that number minus 1, is divisible by 3 or 4 or both. A remarkable discovery is a proposition given by **Iamblichus** in his treatise on Pythagorean philosophy. It is founded on the observation that the Pythagoreans called 1, 10, 100, 1000, units of the first, second, third, fourth 'course' respectively. The theorem is this: If we add any three consecutive numbers, of which the highest is divisible by 3, then add the digits of that sum, then, again, the digits of *that* sum, and so on, the final sum will be 6. Thus, $61 + 62 + 63 = 186$, $1 + 8 + 6 = 15$, $1 + 5 = 6$. This discovery was the more remarkable, because the ordinary Greek numerical symbolism was much less likely to suggest any such property of numbers than our "Arabic" notation would have been.

The works of Nicomachus, Theon of Smyrna, Thymaridas, and others contain at times investigations of subjects which are really algebraic in their nature. Thymaridas in one place uses the Greek word meaning "unknown quantity" in a way which would lead one to believe that algebra was not far distant. Of interest in tracing the invention of algebra are the arithmetical epigrams in the *Palatine Anthology*, which contain about fifty problems leading to linear equations. Before the introduction of algebra these problems were propounded as puzzles. A riddle attributed to Euclid and contained in the *Anthology* is to this effect: A mule and a donkey were walking along, laden with corn. The mule says to the donkey, "If you gave me one measure, I should carry twice as much as you. If I gave you one, we should both carry equal burdens. Tell me their burdens, O most learned master of geometry." [6]

It will be allowed, says Gow, that this problem, if authentic, was not beyond Euclid, and the appeal to geometry smacks of antiquity. A far more difficult puzzle was the famous 'cattle-problem,' which Archimedes propounded to the Alexandrian mathematicians. The problem is indeterminate, for from only seven equations, eight unknown quantities in integral numbers are to be found. It may be stated thus: The sun had a herd of bulls and cows, of different colours. (1) Of Bulls, the white (W) were, in number, $(\frac{1}{2}+\frac{1}{3})$ of the blue (B) and yellow (Y): the B were $(\frac{1}{4}+\frac{1}{5})$ of the Y and piebald (P): the P were $(\frac{1}{6}+\frac{1}{7})$ of the W and Y. (2) Of Cows, which had the same colours (w, b, y, p),

$$\begin{aligned} w = (\tfrac{1}{3}+\tfrac{1}{4})(B+b) : b = (\tfrac{1}{4}+\tfrac{1}{5})(P+p) : p \\ = (\tfrac{1}{5}+\tfrac{1}{6})(Y+y) : y = (\tfrac{1}{6}+\tfrac{1}{7})(W+w). \end{aligned}$$

Find the number of bulls and cows. [6] Another problem in the *Anthology* is quite familiar to school-boys: "Of four pipes, one fills the cistern in one day, the next in two days, the third in three days, the fourth in four days: if all run together, how soon will they fill the cistern?" A great many of these problems, puzzling to an arithmetician, would have been solved easily by an algebraist. They became very popular about the time of Diophantus, and doubtless acted as a powerful stimulus on his mind.

Diophantus was one of the last and most fertile mathematicians of the second Alexandrian school. He died about 330 A.D. His age was eighty-four, as is known from an epitaph to this effect: Diophantus passed $\frac{1}{6}$ of his life in childhood, $\frac{1}{12}$ in youth, and $\frac{1}{7}$ more as a bachelor; five years after his marriage was born a son who died four years before his father, at half his father's age. The place of nativity and parentage of Diophantus are unknown. If his works were not written in Greek, no one would think for a moment that they were the product of Greek mind. There is nothing in his works that reminds us of the classic period of Greek mathematics. His were almost entirely new ideas on a new subject. In the circle of Greek mathematicians he stands alone in his specialty. Except for him, we should be constrained to say that among the Greeks *algebra* was always an unknown science.

Of his works we have lost the *Porisms*, but possess a fragment of *Polygonal Numbers*, and seven books of his great work on *Arithmetica*, said to have been written in 13 books.

If we except the Ahmes papyrus, which contains the first suggestions of algebraic notation, and of the solution of equations, then his *Arithmetica* is the earliest treatise on algebra now extant. In this work is introduced the idea of an algebraic equation expressed in algebraic symbols. His treatment is purely analytical and completely divorced from geometrical methods. He is, as far as we know, the first to state that "a negative number multiplied by a negative number gives a positive number." This is applied to the multiplication of differences, such as $(x-1)(x-2)$. It must be remarked, however, that Diophantus had no notion whatever of negative numbers standing by themselves. All he knew were differences, such as $(2x - 10)$, in which $2x$ could not be smaller than 10 without leading to an absurdity. He appears to be the first who could perform such operations as $(x-1) \times (x-2)$ without reference to geometry. Such identities as $(a+b)^2 = a^2 + 2ab + b^2$, which with Euclid appear in the elevated rank of geometric theorems, are with Diophantus the simplest consequences of the algebraic laws of operation. His sign for subtraction was ⋔, for equality ι. For unknown quantities he had only one symbol, ς. He had no sign for addition except juxtaposition. Diophantus used but few symbols, and sometimes ignored even these by describing an operation in words when the symbol would have answered just as well.

In the solution of simultaneous equations Diophantus adroitly managed with only one symbol for the unknown quantities and arrived at answers, most commonly, by the method of *tentative assumption*, which consists in assigning to some of the unknown quantities preliminary values, that satisfy only one or two of the conditions. These values lead to expressions palpably wrong, but which generally suggest some stratagem by which values can be secured satisfying all the conditions of the problem.

Diophantus also solved determinate equations of the second degree. We are ignorant of his method, for he nowhere goes through with the whole process of solution, but merely states the result. Thus, "$84x^2 + 7x = 7$, whence x is found $= \frac{1}{4}$." Notice he gives only one root. His failure to observe that a quadratic equation has two roots, even when both roots are positive, rather surprises us. It must be remembered, however, that this same inability to perceive more than one out of the several solutions to which a problem may point is common to all Greek mathematicians. Another point to be observed is that he never accepts as an answer a quantity which is negative or irrational.

Diophantus devotes only the first book of his *Arithmetica* to the solution of determinate equations. The remaining books extant treat mainly of *indeterminate quadratic equations* of the form $Ax^2 + Bx + C = y^2$, or of two simultaneous equations of the same form. He considers several but not all the possible cases which may arise in these equations. The opinion of Nesselmann on the method of Diophantus, as stated by Gow, is as follows: "(1) Indeterminate equations of the second degree are treated completely only when the quadratic or the absolute term is wanting: his solution of the equations $Ax^2 + C = y^2$ and $Ax^2 + Bx + C = y^2$ is in many respects cramped. (2) For the 'double equation' of the second degree he has a definite rule only when the quadratic term is wanting in both expressions: even then his solution is not general. More complicated expressions occur only under specially favourable circumstances." Thus, he solves $Bx + C^2 = y^2$, $B_1x + C_1^2 = y_1^2$.

The extraordinary ability of Diophantus lies rather in another direction, namely, in his wonderful ingenuity to reduce all sorts of equations to particular forms which he knows how to solve. Very great is the variety of problems considered. The 130 problems found in the great work of Diophantus contain over 50 different classes of problems, which are strung together without any attempt at classification. But still more multifarious than the problems are the solutions. General methods are unknown to Diophantus. Each problem has its own distinct method, which is often useless for the most closely related problems. "It is, therefore, difficult for a modern, after studying 100 Diophantine solutions, to solve the 101st." [7]

That which robs his work of much of its scientific value is the fact that he always feels satisfied with one solution, though his equation may admit of an indefinite number of values. Another great defect is the absence of general methods. Modern mathematicians, such as Euler, Lagrange, Gauss, had to begin the study of indeterminate analysis anew and received no direct aid from Diophantus in the formulation of methods. In spite of these defects we cannot fail to admire the work for the wonderful ingenuity exhibited therein in the solution of particular equations.

It is still an open question and one of great difficulty whether Diophantus derived portions of his algebra from Hindoo sources or not.

THE ROMANS.

Nowhere is the contrast between the Greek and Roman mind shown forth more distinctly than in their attitude toward the mathematical science. The sway of the Greek was a flowering time for mathematics, but that of the Roman a period of sterility. In philosophy, poetry, and art the Roman was an imitator. But in mathematics he did not even rise to the desire for imitation. The mathematical fruits of Greek genius lay before him untasted. In him a science which had no direct bearing on practical life could awake no interest. As a consequence, not only the higher geometry of Archimedes and Apollonius, but even the *Elements* of Euclid, were entirely neglected. What little mathematics the Romans possessed did not come from the Greeks, but from more ancient sources. Exactly where and how it originated is a matter of doubt. It seems most probable that the "Roman notation," as well as the practical geometry of the Romans, came from the old Etruscans, who, at the earliest period to which our knowledge of them extends, inhabited the district between the Arno and Tiber.

Livy tells us that the Etruscans were in the habit of representing the number of years elapsed, by driving yearly a nail into the sanctuary of Minerva, and that the Romans continued this practice. A less primitive mode of designating numbers, presumably of Etruscan origin, was a notation resembling the present "Roman notation." This system is noteworthy from the fact that a principle is involved in it which is not met with in any other; namely, the principle of subtraction. If a letter be placed before another of greater value, its value is not to be added to, but subtracted from, that of the greater. In the designation of large numbers a horizontal bar placed over a letter was made to increase its value one thousand fold. In fractions the Romans used the duodecimal system.

Of arithmetical calculations, the Romans employed three different kinds: Reckoning on the fingers, upon the abacus, and by tables prepared for the purpose. [3] Finger-symbolism was known as early as the time of King Numa, for he had erected, says Pliny, a statue of the double-faced Janus, of which the fingers indicated 365 (355?), the number of days in a year. Many other passages from Roman authors point out the use of the fingers as aids to calculation. In fact, a finger-symbolism of practically the same form was in use not only in Rome, but also in Greece and throughout the East, certainly as early as the beginning of the Christian era, and continued to be used in Europe during the Middle Ages. We possess no knowledge as to where or when it was invented. The second mode of calculation, by the abacus, was a subject of elementary instruction in Rome. Passages in Roman writers indicate that the kind of abacus most commonly used was covered with dust and then divided into columns by drawing straight lines. Each column was supplied with pebbles (calculi, whence 'calculare' and 'calculate') which served for calculation. Additions and subtractions could be performed on the abacus quite easily, but in multiplication the abacus could be used only for adding the particular products, and in division for performing the subtractions occurring in the process. Doubtless at this point recourse was made to mental operations and to the multiplication table. Possibly finger-multiplication may also have been used. But the multiplication of large numbers must, by either method, have been beyond the power of the ordinary arithmetician. To obviate this difficulty, the arithmetical tables mentioned above were used, from which the desired products could be copied at once. Tables of this kind were prepared by *Victorius* of Aquitania. His tables contain a peculiar notation for fractions, which continued in use throughout the Middle Ages. Victorius is best known for his *canon paschalis*, a rule for finding the correct date for Easter, which he published in 457 A.D.

Payments of interest and problems in interest were very old among the Romans. The Roman laws of inheritance gave rise to numerous arithmetical examples. Especially unique is the following: A dying man wills that, if his wife, being with child, gives birth to a son, the son shall receive $\frac{2}{3}$ and she $\frac{1}{3}$ of his estates; but if a daughter is born, she shall receive $\frac{1}{3}$ and his wife $\frac{2}{3}$. It happens that twins are born, a boy and a girl. How shall the estates be divided so as to satisfy the will? The celebrated Roman jurist, Salvianus Julianus, decided that the estates shall be divided into seven equal parts, of which the son receives four, the wife two, the daughter one.

We next consider Roman geometry. He who expects to find in Rome a science of geometry, with definitions, axioms, theorems, and proofs arranged in logical order, will be disappointed. The only geometry known was a *practical* geometry, which, like the old Egyptian, consisted only of empirical rules. This practical geometry was employed in surveying. Treatises thereon have come down to us, compiled by the Roman surveyors, called *agrimensores* or *gromatici.* One would naturally expect rules to be clearly formulated. But no; they are left to be abstracted by the reader from a mass of numerical examples. "The total impression is as though the Roman gromatic were thousands of years older than Greek geometry, and as though a deluge were lying between the two." Some of their rules were probably inherited from the Etruscans, but others are identical with those of Heron. Among the latter is that for finding the area of a triangle from its sides and the approximate formula, $\frac{13}{30}a^2$, for the area of equilateral triangles (a being one of the sides). But the latter area was also calculated by the formulas $\frac{1}{2}(a^2+a)$ and $\frac{1}{2}a^2$, the first of which was unknown to Heron. Probably the expression $\frac{1}{2}a^2$ was derived from the Egyptian formula $\dfrac{a+b}{2}\cdot\dfrac{c+d}{2}$ for the determination of the surface of a quadrilateral. This Egyptian formula was used by the Romans for finding the area, not only of rectangles, but of any quadrilaterals whatever. Indeed, the gromatici considered it even sufficiently accurate to determine the areas of cities, laid out irregularly, simply by measuring their circumferences. [7] Whatever Egyptian geometry the Romans possessed was transplanted across the Mediterranean at the time of *Julius Cæsar*, who ordered a survey of the whole empire to secure an equitable mode of taxation. Cæsar also reformed the calendar, and, for that purpose, drew from Egyptian learning. He secured the services of the Alexandrian astronomer, *Sosigenes*.

In the fifth century, the Western Roman Empire was fast falling to pieces. Three great branches—Spain, Gaul, and the province of Africa—broke off from the decaying trunk. In 476, the Western Empire passed away, and the Visigothic chief, Odoacer, became king. Soon after, Italy was conquered by the Ostrogoths under Theodoric. It is remarkable that this very period of political humiliation should be the one during which Greek science was studied in Italy most zealously. School-books began to be compiled from the elements of

Greek authors. These compilations are very deficient, but are of absorbing interest, from the fact that, down to the twelfth century, they were the only sources of mathematical knowledge in the Occident. Foremost among these writers is **Boethius** (died 524). At first he was a great favourite of King Theodoric, but later, being charged by envious courtiers with treason, he was imprisoned, and at last decapitated. While in prison he wrote *On the Consolations of Philosophy*. As a mathematician, Boethius was a Brobdingnagian among Roman scholars, but a Liliputian by the side of Greek masters. He wrote an *Institutis Arithmetica*, which is essentially a translation of the arithmetic of Nicomachus, and a *Geometry* in several books. Some of the most beautiful results of Nicomachus are omitted in Boethius' arithmetic. The first book on geometry is an extract from Euclid's *Elements*, which contains, in addition to definitions, postulates, and axioms, the theorems in the first three books, without proofs. How can this omission of proofs be accounted for? It has been argued by some that Boethius possessed an incomplete Greek copy of the *Elements*; by others, that he had Theon's edition before him, and believed that only the theorems came from Euclid, while the proofs were supplied by Theon. The second book, as also other books on geometry attributed to Boethius, teaches, from numerical examples, the mensuration of plane figures after the fashion of the agrimensores.

A celebrated portion in the geometry of Boethius is that pertaining to an abacus, which he attributes to the Pythagoreans. A considerable improvement on the old abacus is there introduced. Pebbles are discarded, and *apices* (probably small cones) are used. Upon each of these apices is drawn a numeral giving it some value below 10. The names of these numerals are pure Arabic, or nearly so, but are added, apparently, by a later hand. These figures are obviously the parents of our modern "Arabic" numerals. The 0 is not mentioned by Boethius in the text. These numerals bear striking resemblance to the Gubar-numerals of the West-Arabs, which are admittedly of Indian origin. These facts have given rise to an endless controversy. Some contended that Pythagoras was in India, and from there brought the nine numerals to Greece, where the Pythagoreans used them secretly. This hypothesis has been generally abandoned, for it is not certain that Pythagoras or any disciple of his ever was in India, nor is there any evidence in any Greek author, that the apices were known to the Greeks, or that numeral signs of any sort were used by them with the abacus. It is improbable, moreover, that the Indian signs, from which the apices are derived, are so old as the time of Pythagoras. A second theory is that the *Geometry* attributed to Boethius is a forgery; that it is not older than the tenth, or possibly the ninth, century, and that the apices are derived from the Arabs. This theory is based on contradictions between passages in the *Arithmetica* and others in the *Geometry*. But there is an Encyclopædia written by *Cassiodorius* (died about 570) in which both the arithmetic and geometry of Boethius are mentioned. There appears to be no good reason for doubting the trustworthiness of this passage in the Encyclopædia. A third theory (Woepcke's) is that the Alexandrians either directly or indirectly obtained the nine numerals from the Hindoos, about the second century A.D., and gave them to the Romans on the one hand, and to the Western Arabs on the other. This explanation is the most plausible.

MIDDLE AGES.

THE HINDOOS.

The first people who distinguished themselves in mathematical research, after the time of the ancient Greeks, belonged, like them, to the Aryan race. It was, however, not a European, but an Asiatic nation, and had its seat in far-off India.

Unlike the Greek, Indian society was fixed into castes. The only castes enjoying the privilege and leisure for advanced study and thinking were the *Brahmins*, whose prime business was religion and philosophy, and the *Kshatriyas*, who attended to war and government.

Of the development of Hindoo mathematics we know but little. A few manuscripts bear testimony that the Indians had climbed to a lofty height, but their path of ascent is no longer traceable. It would seem that Greek mathematics grew up under more favourable conditions than the Hindoo, for in Greece it attained an independent existence, and was studied for its own sake, while Hindoo mathematics always remained merely a servant to astronomy. Furthermore, in Greece mathematics was a science of the people, free to be cultivated by all who had a liking for it; in India, as in Egypt, it was in the hands chiefly of the priests. Again, the Indians were in the habit of putting into verse all mathematical results they obtained, and of clothing them in obscure and mystic language, which, though well adapted to aid the memory of him who already understood the subject, was often unintelligible to the uninitiated. Although the great Hindoo mathematicians doubtless reasoned out most or all of their discoveries, yet they were not in the habit of preserving the proofs, so that the naked theorems and processes of operation are all that have come down to our time. Very different in these respects were the Greeks. Obscurity of language was generally avoided, and proofs belonged to the stock of knowledge quite as much as the theorems themselves. Very striking was the difference in the bent of mind of the Hindoo and Greek; for, while the Greek mind was pre-eminently *geometrical*, the Indian was first of all *arithmetical*. The Hindoo dealt with number, the Greek with form. Numerical symbolism, the science of numbers, and algebra attained in India far greater perfection than they had previously reached in Greece. On the other hand, we believe that there was little or no geometry in India of which the source may not be traced back to Greece. Hindoo trigonometry might possibly be mentioned as an exception, but it rested on arithmetic more than on geometry.

An interesting but difficult task is the tracing of the relation between Hindoo and Greek mathematics. It is well known that more or less trade was carried on between Greece and India from early times. After Egypt had become a Roman province, a more lively commercial intercourse sprang up between Rome and India, by way of Alexandria. *A priori*, it does not seem improbable, that with the traffic of merchandise there should also be an interchange of ideas. That communications of thought from the Hindoos to the Alexandrians actually did take place, is evident from the fact that certain philosophic and theologic teachings of the Manicheans, Neo-Platonists, Gnostics, show unmistakable likeness to Indian tenets. Scientific facts passed also from Alexandria to India. This is shown plainly by the Greek origin of some of the technical terms used by the Hindoos. Hindoo astronomy was influenced by Greek astronomy. Most of the geometrical knowledge which they possessed is traceable to Alexandria, and to the writings of Heron in particular. In algebra there was, probably, a mutual giving and receiving. We suspect that Diophantus got the first glimpses of algebraic knowledge from India. On the other hand, evidences have been found of Greek algebra among the Brahmins. The earliest knowledge of algebra in India may possibly have been of Babylonian origin. When we consider that Hindoo scientists looked upon arithmetic and algebra merely as tools useful in astronomical research, there appears deep irony in the fact that these secondary branches were after all the only ones in which they won real distinction, while in their pet science of astronomy they displayed an inaptitude to observe, to collect facts, and to make inductive investigations.

We shall now proceed to enumerate the names of the leading Hindoo mathematicians, and then to review briefly Indian mathematics. We shall consider the science only in its complete state, for our data are not sufficient to trace the history of the development of methods. Of the great Indian mathematicians, or rather, astronomers,—for India had no mathematicians proper,—**Aryabhatta** is the earliest. He was born 476 A.D., at Pataliputra, on the upper Ganges. His celebrity rests on a work entitled *Aryabhattiyam*, of which the third chapter is devoted to mathematics. About one hundred years later, mathematics in India reached the highest mark. At that time flourished **Brahmagupta** (born 598). In 628 he wrote his *Brahma-sphuta-siddhanta* ("The Revised System of Brahma"), of which the twelfth and eighteenth chapters belong to mathematics. To the fourth or fifth century belongs an anonymous astronomical work, called *Surya-siddhanta* ("Knowledge from the Sun"), which by native authorities was ranked second only to the *Brahma-siddhanta*, but is of interest to us merely as furnishing evidence that Greek science influenced Indian science even before the time of Aryabhatta. The following centuries produced only two names of importance; namely, **Cridhara**, who wrote a *Ganita-sara* ("Quintessence of Calculation"), and **Padmanabha**, the author of an algebra. The science seems to have made but little progress at this time; for a work entitled *Siddhantaciromani* ("Diadem of an Astronomical System"), written by **Bhaskara Acarya** in 1150, stands little higher than that of Brahmagupta, written over 500 years earlier. The two most important mathematical chapters in this work are the *Lilavati* (= "the beautiful," *i.e.* the noble science) and *Viga-ganita* (= "root-extraction"), devoted to arithmetic and algebra. From now on, the Hindoos in the Brahmin schools seemed to content themselves with studying the masterpieces of their predecessors. Scientific intelligence decreases continually, and in modern times a very deficient Arabic work of the sixteenth century has been held in great authority. [7]

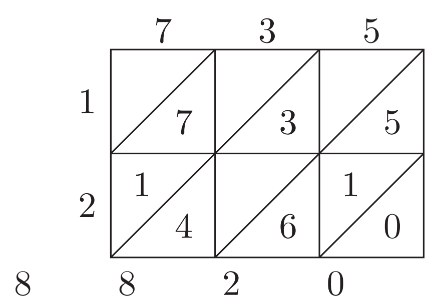
The mathematical chapters of the *Brahma-siddhanta* and *Siddhantaciromani* were translated into English by H. T. Colebrooke, London, 1817. The *Surya-siddhanta* was translated by E. Burgess, and annotated by W. D. Whitney, New Haven, Conn., 1860.

The grandest achievement of the Hindoos and the one which, of all mathematical inventions, has contributed most to the general progress of intelligence, is the invention of the principle of position in writing numbers. Generally we speak of our notation as the "Arabic" notation, but it should be called the "Hindoo" notation, for the Arabs borrowed it from the Hindoos. That the invention of this notation was not so easy as we might suppose at first thought, may be inferred from the fact that, of other nations, not even the keen-minded Greeks possessed one like it. We inquire, who invented this ideal symbolism, and when? But we know neither the inventor nor the time of invention. That our system of notation is of Indian origin is the only point of which we are certain. From the evolution of ideas in general we may safely infer that our notation did not spring into existence a completely armed Minerva from the head of Jupiter. The nine figures for writing the units are supposed to have been introduced earliest, and the sign of zero and the principle of position to be of later origin. This view receives support from the fact that on the island of Ceylon a notation resembling the Hindoo, but without the zero has been preserved. We know that Buddhism and Indian culture were transplanted to Ceylon about the third century after Christ, and that this culture remained stationary there, while it made progress on the continent. It seems highly probable, then, that the numerals of Ceylon are the old, imperfect numerals of India. In Ceylon, nine figures were used for the units, nine others for the tens, one for 100, and also one for 1000. These 20 characters enabled them to write all the numbers up to 9999. Thus, 8725 would have been written with six signs, representing the following numbers: 8, 1000, 7, 100, 20, 5. These Singhalesian signs, like the old Hindoo numerals, are supposed originally to have been the initial letters of the corresponding numeral adjectives. There is a marked resemblance between the notation of Ceylon and the one used by Aryabhatta in the first chapter of his work, and there only. Although the zero and the principle of position were unknown to the scholars of Ceylon, they were probably known to Aryabhatta; for, in the second chapter, he gives directions for extracting the square and cube roots, which seem to indicate a knowledge of them. It would appear that the zero and the accompanying principle of position were introduced about the time of Aryabhatta. These are the inventions which give the Hindoo system its great superiority, its admirable perfection.

There appear to have been several notations in use in different parts of India, which differed, not in principle, but merely in the forms of the signs employed. Of interest is also a *symbolical system of position*, in which the figures generally were not expressed by numerical adjectives, but by objects suggesting the particular numbers in question. Thus, for 1 were used the words *moon*, *Brahma*, *Creator*, or *form*; for 4, the words *Veda*, (because it is divided into four parts) or *ocean*, etc. The following example, taken from the *Surya-siddhanta*, illustrates the idea. The number $1,577,917,828$ is expressed from right to left as follows: Vasu (a class of 8 gods) + two + eight + mountains (the 7 mountain-chains) + form + digits (the 9 digits) + seven + mountains + lunar days (half of which equal 15). The use of such notations made it possible to represent a number in several different ways. This greatly facilitated the framing of verses containing arithmetical rules or scientific constants, which could thus be more easily remembered.

At an early period the Hindoos exhibited great skill in calculating, even with large numbers. Thus, they tell us of an examination to which Buddha, the reformer of the Indian religion, had to submit, when a youth, in order to win the maiden he loved. In arithmetic, after having astonished his examiners by naming all the periods of numbers up to the 53d, he was asked whether he could determine the number of primary atoms which, when placed one against the other, would form a line one mile in length. Buddha found the required answer in this way: 7 primary atoms make a very minute grain of dust, 7 of these make a minute grain of dust, 7 of *these* a grain of dust whirled up by the wind, and so on. Thus he proceeded, step by step, until he finally reached the length of a mile. The multiplication of all the factors gave for the multitude of primary atoms in a mile a number consisting of 15 digits. This problem reminds one of the 'Sand-Counter' of Archimedes.

After the numerical symbolism had been perfected, figuring was made much easier. Many of the Indian modes of operation differ from ours. The Hindoos were generally inclined to follow the motion from left to right, as in writing. Thus, they *added* the left-hand columns first, and made the necessary corrections as they proceeded. For instance, they would have added 254 and 663 thus: $2 + 6 = 8$, $5 + 6 = 11$, which changes 8 into 9, $4 + 3 = 7$. Hence the sum 917. In *subtraction* they had two methods. Thus in 821 – 348 they would say, 8 from $11 = 3$, 4 from $11 = 7$, 3 from $7 = 4$. Or they would say, 8 from $11 = 3$, 5 from $12 = 7$, 4 from $8 = 4$. In *multiplication* of a number by another of only one digit, say 569 by 5, they generally said, $5{\cdot}5 = 25$, $5{\cdot}6 = 30$, which changes 25 into 28, $5{\cdot}9 = 45$, hence the 0 must be increased by 4. The product is 2845. In the multiplication with each other of many-figured numbers, they first multiplied, in the manner just indicated, with the left-hand digit of the multiplier, which was written above the multiplicand, and placed the product above the multiplier. On multiplying with the next digit of the multiplier, the product was not placed in a new row, as with us, but the first product obtained was corrected, as the process continued, by erasing, whenever necessary, the old digits, and replacing them by new ones, until finally the whole product was obtained. We who possess the modern luxuries of pencil and paper, would not be likely to fall in love with this Hindoo method. But the Indians wrote "with a cane-pen upon a small blackboard with a white, thinly liquid paint which made marks that could be easily erased, or upon a white tablet, less than a foot square, strewn with red flour, on which they wrote the figures with a small stick, so that the figures appeared white on a red ground." [7] Since the digits had to be quite large to be distinctly legible, and since the boards were small, it was desirable to have a method which would not require much space. Such a one was the above method of multiplication. Figures could be easily erased and replaced by others without sacrificing neatness. But the Hindoos had also other ways of multiplying, of which we mention the following: The tablet was divided into squares like a chess-board. Diagonals were also drawn, as seen in the figure. The multiplication of $12 \times 735 = 8820$ is exhibited in the adjoining diagram. [3] The manuscripts extant give no information of how *divisions* were executed. The correctness of their additions, subtractions, and multiplications was tested "by excess of 9's." In writing fractions, the numerator was placed above the denominator, but no line was drawn between them.



We shall now proceed to the consideration of some arithmetical problems and the Indian modes of solution. A favourite method was that of *inversion.* With laconic brevity, Aryabhatta describes it thus: "Multiplication becomes division, division becomes multiplication; what was gain becomes loss, what loss, gain; inversion." Quite different from this quotation in style is the following problem from Aryabhatta, which illustrates the method: [3] "Beautiful maiden with beaming eyes, tell me, as thou understandst the right method of inversion, which is the number which multiplied by 3, then increased by $\frac{3}{4}$ of the product, divided by 7, diminished by $\frac{1}{3}$ of the quotient, multiplied by itself, diminished by 52, the square root extracted, addition of 8, and division by 10, gives the number 2?" The process consists in beginning with 2 and working backwards. Thus, $(2{\cdot}10 - 8)^2 + 52 = 196$, $\sqrt{196} = 14$, and $14{\cdot}\frac{3}{2}{\cdot}7{\cdot}\frac{4}{7} \div 3 = 28$, the answer.

Here is another example taken from *Lilavati*, a chapter in Bhaskara's great work: "The square root of half the number of bees in a swarm has flown out upon a jessamine-bush, $\frac{8}{9}$ of the whole swarm has remained behind; one female bee flies about a male that is buzzing within a lotus-flower into which he was allured in the night by its sweet odour, but is now imprisoned in it. Tell me the number of bees." Answer, 72. The pleasing poetic garb in which all arithmetical problems are clothed is due to the Indian practice of writing all school-books in verse, and especially to the fact that these problems, propounded as puzzles, were a favourite social amusement. Says Brahmagupta: "These problems are proposed simply for pleasure; the wise man can invent a thousand others, or he can solve the problems of others by the rules given here. As the sun eclipses the stars by his brilliancy, so the man of knowledge will eclipse the fame of others in assemblies of the people if he proposes algebraic problems, and still more if he solves them."

The Hindoos solved problems in interest, discount, partnership, alligation, summation of arithmetical and geometric series, devised rules for determining the numbers of combinations and permutations, and invented magic squares. It may here be added that chess, the profoundest of all games, had its origin in India.

The Hindoos made frequent use of the "rule of three," and also of the method of "falsa positio," which is almost identical with that of the "tentative assumption" of Diophantus. These and other rules were applied to a large number of problems.

Passing now to *algebra*, we shall first take up the symbols of operation. Addition was indicated simply by juxtaposition as in Diophantine algebra; subtraction, by placing a dot over the subtrahend; multiplication, by putting after the factors, *bha*, the abbreviation of the word *bhavita*, "the product"; division, by placing the divisor beneath the dividend; square-root, by writing *ka*, from the word *karana* (irrational), before the quantity. The unknown quantity was called by Brahmagupta *yâvattâvat* (*quantum tantum*). When several unknown quantities occurred, he gave, unlike Diophantus, to each a distinct name and symbol. The first unknown was designated by the general term "unknown quantity." The rest were distinguished by names of colours, as the black, blue, yellow, red, or green unknown. The initial syllable of each word constituted the symbol for the respective unknown quantity. Thus *yâ* meant x; *kâ* (from *kâlaka* = black) meant y;

yâ kâ bha, "x times y"; *ka* 15 *ka* 10, "$\sqrt{15} - \sqrt{10}$."

The Indians were the first to recognise the existence of absolutely negative quantities. They brought out the difference between positive and negative quantities by attaching to the one the idea of 'possession,' to the other that of 'debts.' The conception also of opposite directions on a line, as an interpretation of + and − quantities, was not foreign to them. They advanced beyond Diophantus in observing that a quadratic has always two roots. Thus Bhaskara gives $x = 50$ and $x = -5$ for the roots of $x^2 - 45x = 250$. "But," says he, "the second value is in this case not to be taken, for it is inadequate; people do not approve of negative roots." Commentators speak of this as if negative roots were seen, but not admitted.

Another important generalisation, says Hankel, was this, that the Hindoos never confined their arithmetical operations to rational numbers. For instance, Bhaskara showed how, by the formula

$$\sqrt{a + \sqrt{b}} = \sqrt{\frac{a + \sqrt{a^2 - b}}{2}} + \sqrt{\frac{a - \sqrt{a^2 - b}}{2}}$$

the square root of the sum of rational and irrational numbers could be found. The Hindoos never discerned the dividing line between numbers and magnitudes, set up by the Greeks, which, though the product of a scientific spirit, greatly retarded the progress of mathematics. They passed from magnitudes to numbers and from numbers to magnitudes without anticipating that gap which to a sharply discriminating mind exists between the continuous and discontinuous.

Yet by doing so the Indians greatly aided the general progress of mathematics. "Indeed, if one understands by algebra the application of arithmetical operations to complex magnitudes of all sorts, whether rational or irrational numbers or space-magnitudes, then the learned Brahmins of Hindostan are the real inventors of algebra." [7]

Let us now examine more closely the Indian algebra. In extracting the square and cube roots they used the formulas $(a+b)^2 = a^2 + 2ab + b^2$ and $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. In this connection Aryabhatta speaks of dividing a number into periods of two and three digits. From this we infer that the principle of position and the zero in the numeral notation were already known to him. In figuring with zeros, a statement of Bhaskara is interesting. A fraction whose denominator is zero, says he, admits of no alteration, though much be added or subtracted. Indeed, in the same way, no change takes place in the infinite and immutable Deity when worlds are destroyed or created, even though numerous orders of beings be taken up or brought forth. Though in this he apparently evinces clear mathematical notions, yet in other places he makes a complete failure in figuring with fractions of zero denominator.

In the Hindoo solutions of determinate equations, Cantor thinks he can see traces of Diophantine methods. Some technical terms betray their Greek origin. Even if it be true that the Indians borrowed from the Greeks, they deserve great credit for improving and generalising the solutions of linear and quadratic equations. Bhaskara advances far beyond the

Greeks and even beyond Brahmagupta when he says that "the square of a positive, as also of a negative number, is positive; that the square root of a positive number is twofold, positive and negative. There is no square root of a negative number, for it is not a square." Of equations of higher degrees, the Indians succeeded in solving only some special cases in which both sides of the equation could be made perfect powers by the addition of certain terms to each.

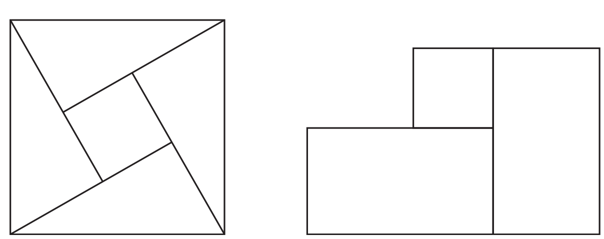
Incomparably greater progress than in the solution of determinate equations was made by the Hindoos in the treatment of *indeterminate equations.* Indeterminate analysis was a subject to which the Hindoo mind showed a happy adaptation. We have seen that this very subject was a favourite with Diophantus, and that his ingenuity was almost inexhaustible in devising solutions for particular cases. But the glory of having invented *general* methods in this most subtle branch of mathematics belongs to the Indians. The Hindoo indeterminate analysis differs from the Greek not only in method, but also in aim. The object of the former was to find all possible integral solutions. Greek analysis, on the other hand, demanded not necessarily integral, but simply rational answers. Diophantus was content with a single solution; the Hindoos endeavoured to find all solutions possible. Aryabhatta gives solutions in integers to linear equations of the form $ax \pm by = c$, where a, b, c are integers. The rule employed is called the *pulveriser.* For this, as for most other rules, the Indians give no proof. Their solution is essentially the same as the one of Euler. Euler's process of reducing $\frac{a}{b}$ to a continued fraction amounts to the same as the Hindoo process of finding the greatest common divisor of a and b by division. This is frequently called the Diophantine method. Hankel protests against this name, on the ground that Diophantus not only never knew the method, but did not even aim at solutions purely integral. [7] These equations probably grew out of problems in astronomy. They were applied, for instance, to determine the time when a certain constellation of the planets would occur in the heavens.

Passing by the subject of linear equations with more than two unknown quantities, we come to indeterminate quadratic equations. In the solution of $xy = ax + by + c$, they applied the method re-invented later by Euler, of decomposing $(ab + c)$ into the product of two integers $m \cdot n$ and of placing $x = m + b$ and $y = n + a$.

Remarkable is the Hindoo solution of the quadratic equation $cy^2 = ax^2 + b$. With great keenness of intellect they recognised in the special case $y^2 = ax^2 + 1$ a fundamental problem in indeterminate quadratics. They solved it by the *cyclic method.* "It consists," says De Morgan, "in a rule for finding an indefinite number of solutions of $y^2 = ax^2 + 1$ (a being an integer which is not a square), by means of one solution given or found, and of feeling for one solution by making a solution of $y^2 = ax^2 + b$ give a solution of $y^2 = ax^2 + b^2$. It amounts to the following theorem: If p and q be one set of values of x and y in $y^2 = ax^2 + b$ and p' and q' the same or another set, then $qp + pq'$ and $app' + qq'$ are values of x and y in $y^2 = ax^2 + b^2$. From this it is obvious that one solution of $y^2 = ax^2 + 1$ may be made to give any number, and that if, taking b at pleasure, $y^2 = ax^2 + b^2$ can be solved so that x and y are divisible by b, then one preliminary solution of $y^2 = ax^2 + 1$ can be found. Another mode of trying for solutions is a combination of the preceding with the *cuttaca* (pulveriser)." These calculations were used in astronomy.

Doubtless this "cyclic method" constitutes the greatest invention in the theory of numbers before the time of Lagrange. The perversity of fate has willed it, that the equation $y^2 = ax^2 + 1$ should now be called *Pell's* problem, while in recognition of Brahmin scholarship it ought to be called the "Hindoo problem." It is a problem that has exercised the highest faculties of some of our greatest modern analysts. By them the work of the Hindoos was done over again; for, unfortunately, the Arabs transmitted to Europe only a small part of Indian algebra and the original Hindoo manuscripts, which we now possess, were unknown in the Occident.

Hindoo *geometry* is far inferior to the Greek. In it are found no definitions, no postulates, no axioms, no logical chain of reasoning or rigid form of demonstration, as with Euclid. Each theorem stands by itself as an independent truth. Like the early Egyptian, it is empirical. Thus, in the proof of the theorem of the right triangle, Bhaskara draws the right triangle four times in the square of the hypotenuse, so that in the middle there remains a square whose side equals



the difference between the two sides of the right triangle. Arranging this square and the four triangles in a different way, they are seen, together, to make up the sum of the square of the two sides. "Behold!" says Bhaskara, without adding another word of explanation. Bretschneider conjectures that the Pythagorean proof was substantially the same as this. In another place, Bhaskara gives a second demonstration of this theorem by drawing from the vertex of the right angle a perpendicular to the hypotenuse, and comparing the two triangles thus obtained with the given triangle to which they are similar. This proof was unknown in Europe till Wallis rediscovered it. The Brahmins never inquired into the properties of figures. They considered only metrical relations applicable in practical life. In the Greek sense, the Brahmins never had a science of geometry. Of interest is the formula given by Brahmagupta for the area of a triangle in terms of its sides. In the great work attributed to Heron the Elder this formula is first found. Whether the Indians themselves invented it, or whether they borrowed it from Heron, is a disputed question. Several theorems are given by Brahmagupta on quadrilaterals which are true only of those which can be inscribed on a circle—a limitation which he omits to state. Among these is the proposition of Ptolemæus, that the product of the diagonals is equal to the sum of the products of the opposite sides. The Hindoos were familiar with the calculation of the areas of circles and their segments, of the length of chords and perimeters of regular inscribed polygons. An old Indian tradition makes $\pi = 3$, also $= \sqrt{10}$; but Aryabhatta gives the value $\frac{31416}{10000}$. Bhaskara gives two values,—the 'accurate,' $\frac{3927}{1250}$, and the 'inaccurate,' Archimedean value, $\frac{22}{7}$. A commentator on *Lilavati* says that these values were calculated by beginning with a regular inscribed hexagon, and applying repeatedly the formula $AD = \sqrt{2 - \sqrt{4 - \overline{AB}^2}}$, wherein AB is the side of the given polygon, and AD that of one with double the number of sides. In this way were obtained the perimeters of the inscribed polygons of 12, 24, 48, 96, 192, 384 sides. Taking the radius $= 100$, the perimeter of the last one gives the value which Aryabhatta used for π.

Greater taste than for geometry was shown by the Hindoos for *trigonometry*. Like the Babylonians and Greeks, they divided the circle into quadrants, each quadrant into 90 degrees and 5400 minutes. The whole circle was therefore made up of $21,600$ equal parts. From Bhaskara's 'accurate' value for π it was found that the radius contained 3438 of these circular parts. This last step was not Grecian. The Greeks might have had scruples about taking a part of a *curve* as the measure of a *straight line*. Each quadrant was divided into 24 equal parts, so that each part embraced 225 units of the whole circumference, and corresponds to $3\frac{3}{4}$ degrees. Notable is the fact that the Indians never reckoned, like the Greeks, with the whole chord of double the arc, but always with the *sine* (*joa*) and *versed sine*. Their mode of calculating tables was theoretically very simple. The sine of $90°$ was equal to the radius, or 3438; the sine of $30°$ was evidently half that, or 1719. Applying the formula $\sin^2 a + \cos^2 a = r^2$, they obtained $\sin 45^\circ = \sqrt{\frac{r^2}{2}} = 2431$. Substituting for $\cos a$ its equal $\sin(90 - a)$, and making $a = 60^\circ$, they obtained $\sin 60^\circ = \frac{\sqrt{3r^2}}{2} = 2978$. With the sines of 90, 60, 45, and 30 as starting-points, they reckoned the sines of half the angles by the formula $\operatorname{versin} 2a = 2\sin^2 a$, thus obtaining the sines of $22^\circ\ 30'$, $11^\circ\ 15'$, $7^\circ\ 30'$, $3^\circ\ 45'$. They now figured out the sines of the complements of these angles, namely, the sines of $86^\circ\ 15'$, $82^\circ\ 30'$, $78^\circ\ 45'$, 75°, $67^\circ\ 30'$; then they calculated the sines of half these angles; then of their complements; then, again, of half their complements; and so on. By this very simple process they got the sines of angles at intervals of $3^\circ\ 45'$. In this table they discovered the unique law that if a, b, c be three successive arcs such that $a - b = b - c = 3^\circ\ 45'$, then $\sin a - \sin b = (\sin b - \sin c) - \frac{\sin b}{225}$. This formula was afterwards used whenever a re-calculation of tables had to be made. No Indian trigonometrical treatise on the triangle is extant. In astronomy they solved plane and spherical right triangles. [18]

It is remarkable to what extent Indian mathematics enters into the science of our time. Both the form and the spirit of the arithmetic and algebra of modern times are essentially Indian and not Grecian. Think of that most perfect of mathematical symbolisms—the Hindoo notation, think of the Indian arithmetical operations nearly as perfect as our own, think of their elegant algebraical methods, and then judge whether the Brahmins on the banks of the Ganges are not entitled to some credit. Unfortunately, some of the most brilliant of Hindoo discoveries in indeterminate analysis reached Europe too late to exert the influence they would have exerted, had they come two or three centuries earlier.

THE ARABS.

After the flight of Mohammed from Mecca to Medina in 622 A.D., an obscure people of Semitic race began to play an important part in the drama of history. Before the lapse of ten years, the scattered tribes of the Arabian peninsula were fused by the furnace blast of religious enthusiasm into a powerful nation. With sword in hand the united Arabs subdued Syria and Mesopotamia. Distant Persia and the lands beyond, even unto India, were added to the dominions of the Saracens. They conquered Northern Africa, and nearly the whole Spanish peninsula, but were finally checked from further progress in Western Europe by the firm hand of Charles Martel (732 A.D.). The Moslem dominion extended now from India to Spain; but a war of succession to the caliphate ensued, and in 755 the Mohammedan empire was divided,—one caliph reigning at Bagdad, the other at Cordova in Spain. Astounding as was the grand march of conquest by the Arabs, still more so was the ease with which they put aside their former nomadic life, adopted a higher civilisation, and assumed the sovereignty over cultivated peoples. Arabic was made the written language throughout the conquered lands. With the rule of the Abbasides in the East began a new period in the history of learning. The capital, Bagdad, situated on the Euphrates, lay half-way between two old centres of scientific thought,—India in the East, and Greece in the West. The Arabs were destined to be the custodians of the torch of Greek and Indian science, to keep it ablaze during the period of confusion and chaos in the Occident, and afterwards to pass it over to the Europeans. Thus science passed from Aryan to Semitic races, and then back again to the Aryan. The Mohammedans have added but little to the knowledge in mathematics which they received. They now and then explored a small region to which the path had been previously pointed out, but they were quite incapable of discovering new fields. Even the more elevated regions in which the Hellenes and Hindoos delighted to wander—namely, the Greek conic sections and the Indian indeterminate analysis—were seldom entered upon by the Arabs. They were less of a speculative, and more of a practical turn of mind.

The Abbasides at Bagdad encouraged the introduction of the sciences by inviting able specialists to their court, irrespective of nationality or religious belief. Medicine and astronomy were their favourite sciences. Thus Haroun-al-Raschid, the most distinguished Saracen ruler, drew Indian physicians to Bagdad. In the year 772 there came to the court of Caliph Almansur a Hindoo astronomer with astronomical tables which were ordered to be translated into Arabic. These tables, known by the Arabs as the *Sindhind*, and probably taken from the *Brahma-sphuta-siddhanta* of Brahmagupta, stood in great authority. They contained the important Hindoo table of sines.

Doubtless at this time, and along with these astronomical tables, the Hindoo numerals, with the zero and the principle of position, were introduced among the Saracens. Before the time of Mohammed the Arabs had no numerals. Numbers were written out in words. Later, the numerous computations connected with the financial administration over the conquered lands made a short symbolism indispensable. In some localities, the numerals of the more civilised conquered nations were used for a time. Thus in Syria, the Greek notation was retained; in Egypt, the Coptic. In some cases, the numeral adjectives may have been abbreviated in writing. The *Diwani-numerals*, found in an Arabic-Persian dictionary, are supposed to be such abbreviations. Gradually it became the practice to employ the 28 Arabic letters of the alphabet for numerals, in analogy to the Greek system. This notation was in turn superseded by the Hindoo notation, which quite early was adopted by merchants, and also by writers on arithmetic. Its superiority was so universally recognised, that it had no rival, except in astronomy, where the alphabetic notation continued to be used. Here the alphabetic notation offered no great disadvantage, since in the sexagesimal arithmetic, taken from the *Almagest*, numbers of generally only one or two places had to be written. [7]

As regards the form of the so-called Arabic numerals, the statement of the Arabic writer *Albiruni* (died 1039), who spent many years in India, is of interest. He says that the shape of the numerals, as also of the letters in India, differed in different localities, and that the Arabs selected from the various forms the most suitable. An Arabian astronomer says there was among people much difference in the use of symbols, especially of those for 5, 6, 7, and 8. The symbols used by the Arabs can be traced back to the tenth century. We find material differences between those used by the Saracens in the East and those used in the West. But most surprising is the fact that the symbols of both the East and of the West Arabs deviate so extraordinarily from the Hindoo *Devanagari* numerals (= divine numerals) of to-day, and that they resemble much more closely the apices of the Roman writer Boethius. This strange similarity on the one hand, and dissimilarity on the other, is difficult to explain. The most plausible theory is the one of Woepcke: (1) that about the second century after Christ, before the zero had been invented, the Indian numerals were brought to Alexandria, whence they spread to Rome and also to West Africa; (2) that in the eighth century, after the notation in India had been already much modified and perfected by the invention of the zero, the Arabs at Bagdad got it from the Hindoos; (3) that the Arabs of the West borrowed the Columbus-egg, the zero, from those in the East, but retained the old forms of the nine numerals, if for no other reason, simply to be contrary to their political enemies of the East; (4) that the old forms were remembered by the West-Arabs to be of Indian origin, and were hence called *Gubar-numerals* (= dust-numerals, in memory of the Brahmin practice of reckoning on tablets strewn with dust or sand; (5) that, since the eighth century, the numerals in India underwent further changes, and assumed the greatly modified forms of the modern Devanagari-numerals. [3] This is rather a bold theory, but, whether true or not, it explains better than any other yet propounded, the relations between the apices, the Gubar, the East-Arabic, and Devanagari numerals.

It has been mentioned that in 772 the Indian *Siddhanta* was brought to Bagdad and there translated into Arabic. There is no evidence that any intercourse existed between Arabic and Indian astronomers either before or after this time, excepting the travels of Albiruni. But we should be very slow to deny the probability that more extended communications actually did take place.

Better informed are we regarding the way in which Greek science, in successive waves, dashed upon and penetrated Arabic soil. In Syria the sciences, especially philosophy and medicine, were cultivated by Greek Christians. Celebrated were the schools at Antioch and Emesa, and, first of all, the flourishing Nestorian school at Edessa. From Syria, Greek physicians and scholars were called to Bagdad. Translations of works from the Greek began to be made. A large number of Greek manuscripts were secured by Caliph *Al Mamun* (813–833) from the emperor in Constantinople and were turned over to Syria. The successors of Al Mamun continued the work so auspiciously begun, until, at the beginning of the tenth century, the more important philosophic, medical, mathematical, and astronomical works of the Greeks could all be read in the Arabic tongue. The translations of mathematical works must have been very deficient at first, as it was evidently difficult to secure translators who were masters of both the Greek and Arabic and at the same time proficient in mathematics. The translations had to be revised again and again before they were satisfactory. The first Greek authors made to speak in Arabic were Euclid and Ptolemæus. This was accomplished during the reign of the famous Haroun-al-Raschid. A revised translation of Euclid's *Elements* was ordered by Al Mamun. As this revision still contained numerous errors, a new translation was made, either by the learned Honein ben Ishak, or by his son, Ishak ben Honein. To the thirteen books of the *Elements* were added the fourteenth, written by Hypsicles, and the fifteenth by Damascius. But it remained for Tabit ben Korra to bring forth an Arabic Euclid satisfying every need. Still greater difficulty was experienced in securing an intelligible translation of the *Almagest*. Among other important translations into Arabic were the works of Apollonius, Archimedes, Heron, and Diophantus. Thus we see that in the course of one century the Arabs gained access to the vast treasures of Greek science. Having been little accustomed to abstract thought, we need not marvel if, during the ninth century, all their energy was exhausted merely in appropriating the foreign material. No attempts were made at original work in mathematics until the next century.

In astronomy, on the other hand, great activity in original research existed as early as the ninth century. The religious observances demanded by Mohammedanism presented to astronomers several practical problems. The Moslem dominions being of such enormous extent, it remained in some localities for the astronomer to determine which way the "Believer" must turn during prayer that he may be facing Mecca. The prayers and ablutions had to take place at definite hours during the day and night. This led to more accurate determinations of time. To fix the exact date for the Mohammedan feasts it became necessary to observe more closely the motions of the moon. In addition to all this, the old Oriental superstition that extraordinary occurrences in the heavens in some mysterious way affect the progress of human affairs added increased interest to the prediction of eclipses. [7]

For these reasons considerable progress was made. Astronomical tables and instruments were perfected, observatories erected, and a connected series of observations instituted. This intense love for astronomy and astrology continued during the whole Arabic scientific period. As in India, so here, we hardly ever find a man exclusively devoted to pure mathematics. Most of the so-called mathematicians were first of all astronomers.

The first notable author of mathematical books was **Mohammed ben Musa Al Hovarezmi**, who lived during the reign of Caliph Al Mamun (813–833). He was engaged by the caliph in making extracts from the *Sindhind*, in revising the tablets of Ptolemæus, in taking observations at Bagdad and Damascus, and in measuring a degree of the earth's meridian. Important to us is his work on algebra and arithmetic. The portion on arithmetic is not extant in the original, and it was not till 1857 that a Latin translation of it was found. It begins thus: "Spoken has Algoritmi. Let us give deserved praise to God, our leader and defender." Here the name of the author, *Al Hovarezmi*, has passed into *Algoritmi*, from which comes our modern word, *algorithm*, signifying the art of computing in any particular way. The arithmetic of Hovarezmi, being based on the principle of position and the Hindoo method of calculation, "excels," says an Arabic writer, "all others in brevity and easiness, and exhibits the Hindoo intellect and sagacity in the grandest inventions." This book was followed by a large number of arithmetics by later authors, which differed from the earlier ones chiefly in the greater variety of methods. Arabian arithmetics generally contained the four operations with integers and fractions, modelled after the Indian processes. They explained the operation of *casting out the 9's*, which was sometimes called the "Hindoo proof." They contained also the *regula falsa* and the *regula duorum falsorum*, by which algebraical examples could be solved without algebra. Both these methods were known to the Indians. The *regula falsa* or *falsa positio* was the assigning of an assumed value to the unknown quantity, which value, if wrong, was corrected by some process like the "rule of three." Diophantus used a method almost identical with this. The *regula duorum falsorum* was as follows: [7] To solve an equation $f(x) = V$, assume, for the moment, two values for x; namely, $x = a$ and $x = b$. Then form $f(a) = A$ and $f(b) = B$, and determine the errors $V - A = E_a$ and $V - B = E_b$; then the required $x = \dfrac{bE_a - aE_b}{E_a - E_b}$ is generally a close approximation, but is absolutely accurate whenever $f(x)$ is a linear function of x.

We now return to Hovarezmi, and consider the other part of his work,—the *algebra*. This is the first book known to contain this word itself as title. Really the title consists of two words, *aldshebr walmukabala*, the nearest English translation of which is "restoration" and "reduction." By "restoration" was meant the transposing of negative terms to the other side of the equation; by "reduction," the uniting of similar terms. Thus, $x^2 - 2x = 5x + 6$ passes by aldshebr into $x^2 = 5x + 2x + 6$; and this, by walmukabala, into $x^2 = 7x + 6$. The work on algebra, like the arithmetic, by the same author, contains nothing original. It explains the elementary operations and the solutions of linear and quadratic equations. From whom did the author borrow his knowledge of algebra? That it came entirely from Indian sources is impossible, for the Hindoos had no such rules like the "restoration" and "reduction." They were, for instance, never in the habit of making all terms in an equation positive, as is done by the process of "restoration." Diophantus gives two rules which resemble somewhat those of our Arabic author, but the probability that the Arab got all his algebra from Diophantus is lessened by the considerations that he recognised both roots of a quadratic, while Diophantus noticed only one; and that the Greek algebraist, unlike the Arab, habitually rejected irrational solutions. It would seem, therefore, that the algebra of Hovarezmi was neither purely Indian nor purely Greek, but was a hybrid of the two, with the Greek element predominating.

The algebra of Hovarezmi contains also a few meagre fragments on *geometry*. He gives the theorem of the right triangle, but proves it after Hindoo fashion and only for the simplest case, when the right triangle is isosceles. He then calculates the areas of the triangle, parallelogram, and circle. For π he uses the value $3\frac{1}{7}$, and also the two Indian, $\pi = \sqrt{10}$ and $\pi = \frac{62832}{20000}$. Strange to say, the last value was afterwards forgotten by the Arabs, and replaced by others less accurate. This bit of geometry doubtless came from India. Later Arabic writers got their geometry almost entirely from Greece.

Next to be noticed are the three sons of **Musa ben Sakir**, who lived in Bagdad at the court of the Caliph Al Mamun. They wrote several works, of which we mention a geometry in which is also contained the well-known formula for the area of a triangle expressed in terms of its sides. We are told that one of the sons travelled to Greece, probably to collect astronomical and mathematical manuscripts, and that on his way back he made acquaintance with Tabit ben Korra. Recognising in him a talented and learned astronomer, Mohammed procured for him a place among the astronomers at the court in Bagdad. **Tabit ben Korra** (836–901) was born at Harran in Mesopotamia. He was proficient not only in astronomy and mathematics, but also in the Greek, Arabic, and Syrian languages. His translations of Apollonius, Archimedes, Euclid, Ptolemy, Theodosius, rank among the best. His dissertation on *amicable numbers* (of which each is the sum of the factors of the other) is the first known specimen of original work in mathematics on Arabic soil. It shows that he was familiar with the Pythagorean theory of numbers. Tabit invented the following rule for finding amicable numbers: If $p = 3{\cdot}2^n - 1$, $q = 3{\cdot}2^{n-1} - 1$, $r = 9{\cdot}2^{2n-1} - 1$ (n being a whole number) are three primes, then $a = 2^n pq$, $b = 2^n r$ are a pair of amicable numbers. Thus, if $n = 2$, then $p = 11$, $q = 5$, $r = 71$, and $a = 220$, $b = 284$. Tabit also trisected an angle.

Foremost among the astronomers of the ninth century ranked **Al Battani**, called *Albategnius* by the Latins. Battan in Syria was his birthplace. His observations were celebrated for great precision. His work, *De scientia stellarum*, was translated into Latin by Plato Tiburtinus, in the twelfth century. Out of this translation sprang the word 'sinus,' as the name of a trigonometric function. The Arabic word for "sine," *dschiba*, was derived from the Sanscrit *jiva*, and resembled the Arabic word *dschaib*, meaning an indentation or gulf. Hence the Latin "sinus." [3] Al Battani was a close student of Ptolemy, but did not follow him altogether. He took an important step for the better, when he introduced the Indian "sine" or *half* the chord, in place of the *whole* chord of Ptolemy. Another improvement on Greek trigonometry made by the Arabs points likewise to Indian influences. Propositions and operations which were treated by the Greeks geometrically are expressed by the Arabs algebraically. Thus, *Al Battani* at once gets from an equation $\dfrac{\sin\theta}{\cos\theta} = D$, the value of θ by means of $\sin\theta = \dfrac{D}{\sqrt{1+D^2}}$,—a process unknown to the ancients. He knows, of course, all the formulas for spherical triangles given in the *Almagest*, but goes further, and adds an important one of his own for oblique-angled triangles; namely, $\cos a = \cos b \cos c + \sin b \sin c \cos A$.

At the beginning of the tenth century political troubles arose in the East, and as a result the house of the Abbasides lost power. One province after another was taken, till, in 945, all possessions were wrested from them. Fortunately, the new rulers at Bagdad, the Persian Buyides, were as much interested in astronomy as their predecessors. The progress of the sciences was not only unchecked, but the conditions for it became even more favourable. The Emir *Adud-ed-daula* (978–983) gloried in having studied astronomy himself. His son *Saraf-ed-daula* erected an observatory in the garden of his palace, and called thither a whole group of scholars. [7] Among them were *Abul Wefa*, *Al Kuhi*, *Al Sagani*.

Abul Wefa (940–998) was born at Buzshan in Chorassan, a region among the Persian mountains, which has brought forth many Arabic astronomers. He forms an important exception to the unprogressive spirit of Arabian scientists by his brilliant discovery of the *variation* of the moon, an inequality usually supposed to have been first discovered by Tycho Brahe. [11] Abul Wefa translated Diophantus. He is one of the last Arabic translators and commentators of Greek authors. The fact that he esteemed the algebra of Mohammed ben Musa Hovarezmi worthy of his commentary indicates that thus far algebra had made little or no progress on Arabic soil. Abul Wefa invented a method for computing tables of sines which gives the sine of half a degree correct to nine decimal places. He did himself credit by introducing the *tangent* into trigonometry and by calculating a table of tangents. The first step toward this had been taken by Al Battani. Unfortunately, this innovation and the discovery of the moon's variation excited apparently no notice among his contemporaries and followers. "We can hardly help looking upon this circumstance as an evidence of a servility of intellect belonging to the Arabian period." A treatise by Abul Wefa on "geometric constructions" indicates that efforts were being made at that time to improve draughting. It contains a neat construction of the corners of the regular polyhedrons on the circumscribed sphere. Here, for the first time, appears the condition which afterwards became very famous in the Occident, that the construction be effected with a single opening of the compass.

Al Kuhi, the second astronomer at the observatory of the emir at Bagdad, was a close student of Archimedes and Apollonius. He solved the problem, to construct a segment of a sphere equal in volume to a given segment and having a curved surface equal in area to that of another given segment. He, **Al Sagani**, and **Al Biruni** made a study of the trisection of angles. **Abul Gud**, an able geometer, solved the problem by the intersection of a parabola with an equilateral hyperbola.

The Arabs had already discovered the theorem that the sum of two cubes can never be a cube. **Abu Mohammed Al Hogendi** of Chorassan thought he had proved this, but we are told that the demonstration was defective. Creditable work in theory of numbers and algebra was done by **Al Karhi** of Bagdad, who lived at the beginning of the eleventh century. His treatise on algebra is the greatest algebraic work of the Arabs. In it he appears as a disciple of Diophantus. He was the first to operate with higher roots and to solve equations of the form $x^{2n} + ax^n = b$. For the solution of quadratic equations he gives both arithmetical and geometric proofs. He was the first Arabic author to give and prove the theorems on the summation of the series:—

$$1^2 + 2^2 + 3^2 + \cdots + n^2 = (1 + 2 + \cdots + n)\,\frac{2n+1}{3},$$
$$1^3 + 2^3 + 3^3 + \cdots + n^3 = (1 + 2 + \cdots + n)^2.$$

Al Karhi also busied himself with indeterminate analysis. He showed skill in handling the methods of Diophantus, but added nothing whatever to the stock of knowledge already on hand. As a subject for original research, indeterminate analysis was too subtle for even the most gifted of Arabian minds. Rather surprising is the fact that Al Karhi's algebra shows no traces whatever of Hindoo indeterminate analysis. But most astonishing it is, that an arithmetic by the same author completely excludes the Hindoo numerals. It is constructed wholly after Greek pattern. Abul Wefa also, in the second half of the tenth century, wrote an arithmetic in which Hindoo numerals find no place. This practice is the very opposite to that of other Arabian authors. The question, why the Hindoo numerals were ignored by so eminent authors, is certainly a puzzle. Cantor suggests that at one time there may have been rival schools, of which one followed almost exclusively Greek mathematics, the other Indian.

The Arabs were familiar with geometric solutions of quadratic equations. Attempts were now made to solve cubic equations geometrically. They were led to such solutions by the study of questions like the Archimedean problem, demanding the section of a sphere by a plane so that the two segments shall be in a prescribed ratio. The first to state this problem in form of a cubic equation was **Al Mahani** of Bagdad, while **Abu Gafar Al Hazin** was the first Arab to solve the equation by conic sections. Solutions were given also by Al Kuhi, Al Hasan ben Al Haitam, and others. [20] Another difficult problem, to determine the side of a regular heptagon, required the construction of the side from the equation $x^3 - x^2 - 2x + 1 = 0$. It was attempted by many and at last solved by Abul Gud.

The one who did most to elevate to a *method* the solution of algebraic equations by intersecting conics, was **Omar al Hayyami** of Chorassan, about 1079 A.D. He divides cubics into two classes, the trinomial and quadrinomial, and each class into families and species. Each species is treated separately but according to a general plan. He believed that cubics could not be solved by calculation, nor biquadratics by geometry. He rejected negative roots and often failed to discover all the positive ones. Attempts at biquadratic equations were made by Abul Wefa, [20] who solved geometrically $x^4 = a$ and $x^4 + ax^3 = b$.

The solution of cubic equations by intersecting conics was the greatest achievement of the Arabs in algebra. The foundation to this work had been laid by the Greeks, for it was Menæchmus who first constructed the roots of $x^3 - a = 0$ or $x^3 - 2a^3 = 0$. It was not his aim to find the number corresponding to x, but simply to determine the side x of a cube double another cube of side a. The Arabs, on the other hand, had another object in view: to find the roots of given numerical equations. In the Occident, the Arabic solutions of cubics remained unknown until quite recently. Descartes and Thomas Baker invented these constructions anew. The works of Al Hayyami, Al Karhi, Abul Gud, show how the Arabs departed further and further from the Indian methods, and placed themselves more immediately under Greek influences. In this way they barred the road of progress against themselves. The Greeks had advanced to a point where material progress became difficult with their methods; but the Hindoos furnished new ideas, many of which the Arabs now rejected.

With Al Karhi and Omar Al Hayyami, mathematics among the Arabs of the East reached flood-mark, and now it begins to ebb. Between 1100 and 1300 A.D. come the crusades with war and bloodshed, during which European Christians profited much by their contact with Arabian culture, then far superior to their own; but the Arabs got no science from the Christians in return. The crusaders were not the only adversaries of the Arabs. During the first half of the thirteenth century, they had to encounter the wild Mongolian hordes, and, in 1256, were conquered by them under the leadership of *Hulagu*. The caliphate at Bagdad now ceased to exist. At the close of the fourteenth century still another empire was formed by Timur or *Tamerlane*, the Tartar. During such sweeping turmoil, it is not surprising that science declined. Indeed, it is a marvel that it existed at all. During the supremacy of Hulagu, lived **Nasir Eddin** (1201–1274), a man of broad culture and an able astronomer. He persuaded Hulagu to build him and his associates a large observatory at Maraga. Treatises on algebra, geometry, arithmetic, and a translation of Euclid's *Elements*, were prepared by him. Even at the court of Tamerlane in Samarkand, the sciences were by no means neglected. A group of astronomers was drawn to this court. **Ulug Beg** (1393–1449), a grandson of Tamerlane, was himself an astronomer. Most prominent at this time was **Al Kaschi**, the author of an arithmetic. Thus, during intervals of peace, science continued to be cultivated in the East for several centuries. The last Oriental writer was *Beha Eddin* (1547–1622). His *Essence of Arithmetic* stands on about the same level as the work of Mohammed ben Musa Hovarezmi, written nearly 800 years before.

"Wonderful is the expansive power of Oriental peoples, with which upon the wings of the wind they conquer half the world, but more wonderful the energy with which, in less than two generations, they raise themselves from the lowest stages of cultivation to scientific efforts." During all these centuries, astronomy and mathematics in the Orient greatly excel these sciences in the Occident.

Thus far we have spoken only of the Arabs in the East. Between the Arabs of the East and of the West, which were under separate governments, there generally existed considerable political animosity. In consequence of this, and of the enormous distance between the two great centres of learning, Bagdad and Cordova, there was less scientific intercourse among them than might be expected to exist between peoples having the same religion and written language. Thus the course of science in Spain was quite independent of that in Persia. While wending our way westward to Cordova, we must stop in Egypt long enough to observe that there, too, scientific activity was rekindled. Not Alexandria, but Cairo with its library and observatory, was now the home of learning. Foremost among her scientists ranked **Ben Junus** (died 1008), a contemporary of Abul Wefa. He solved some difficult problems in spherical trigonometry. Another Egyptian astronomer was **Ibn Al Haitam** (died 1038), who wrote on geometric loci. Travelling westward, we meet in Morocco **Abul Hasan Ali**, whose treatise 'on astronomical instruments' discloses a thorough knowledge of the *Conics* of Apollonius. Arriving finally in Spain at the capital, Cordova, we are struck by the magnificent splendour of her architecture. At this renowned seat of learning, schools and libraries were founded during the tenth century.

Little is known of the progress of mathematics in Spain. The earliest name that has come down to us is **Al Madshriti** (died 1007), the author of a mystic paper on 'amicable numbers.' His pupils founded schools at Cordova, Dania, and Granada. But the only great astronomer among the Saracens in Spain is **Gabir ben Aflah** of Sevilla, frequently called *Geber.* He lived in the second half of the eleventh century. It was formerly believed that he was the inventor of algebra, and that the word *algebra* came from 'Gabir' or 'Geber.' He ranks among the most eminent astronomers of this time, but, like so many of his contemporaries, his writings contain a great deal of mysticism. His chief work is an astronomy in nine books, of which the first is devoted to trigonometry. In his treatment of spherical trigonometry, he exercises great independence of thought. He makes war against the time-honoured procedure adopted by Ptolemy of applying "the rule of six quantities," and gives a new way of his own, based on the 'rule of four quantities.' This is: If PP_1 and QQ_1 be two arcs of great circles intersecting in A, and if PQ and P_1Q_1 be arcs of great circles drawn perpendicular to QQ_1, then we have the proportion

$$\sin AP : \sin PQ = \sin AP_1 : \sin P_1Q_1.$$

From this he derives the formulas for spherical right triangles. To the four fundamental formulas already given by Ptolemy, he added a fifth, discovered by himself. If a, b, c, be the sides, and A, B, C, the angles of a spherical triangle, right-angled at A, then $\cos B = \cos b \sin C$. This is frequently called "Geber's Theorem." Radical and bold as were his innovations in spherical trigonometry, in plane trigonometry he followed slavishly the old beaten path of the Greeks. Not even did he adopt the Indian 'sine' and 'cosine,' but still used the Greek 'chord of double the angle.' So painful was the departure from old ideas, even to an independent Arab! After the time of Gabir ben Aflah there was no mathematician among the Spanish Saracens of any reputation. In the year in which Columbus discovered America, the Moors lost their last foothold on Spanish soil.

We have witnessed a laudable intellectual activity among the Arabs. They had the good fortune to possess rulers who, by their munificence, furthered scientific research. At the courts of the caliphs, scientists were supplied with libraries and observatories. A large number of astronomical and mathematical works were written by Arabic authors. Yet we fail to find a single important principle in mathematics brought forth by the Arabic mind. Whatever discoveries they made, were in fields previously traversed by the Greeks or the Indians, and consisted of objects which the latter had overlooked in their rapid march. The Arabic mind did not possess that penetrative insight and invention by which mathematicians in Europe afterwards revolutionised the science. The Arabs were learned, but not original. Their chief service to science consists in this, that they adopted the learning of Greece and India, and kept what they received with scrupulous care. When the love for science began to grow in the Occident, they transmitted to the Europeans the valuable treasures of antiquity. Thus a Semitic race was, during the Dark Ages, the custodian of the Aryan intellectual possessions.

EUROPE DURING THE MIDDLE AGES.

With the third century after Christ begins an era of migration of nations in Europe. The powerful Goths quit their swamps and forests in the North and sweep onward in steady southwestern current, dislodging the Vandals, Sueves, and Burgundians, crossing the Roman territory, and stopping and recoiling only when reaching the shores of the Mediterranean. From the Ural Mountains wild hordes sweep down on the Danube. The Roman Empire falls to pieces, and the Dark Ages begin. But dark though they seem, they are the germinating season of the institutions and nations of modern Europe. The Teutonic element, partly pure, partly intermixed with the Celtic and Latin, produces that strong and luxuriant growth, the modern civilisation of Europe. Almost all the various nations of Europe belong to the Aryan stock. As the Greeks and the Hindoos—both Aryan races—were the great thinkers of antiquity, so the nations north of the Alps became the great intellectual leaders of modern times.

Introduction of Roman Mathematics.

We shall now consider how these as yet barbaric nations of the North gradually came in possession of the intellectual treasures of antiquity. With the spread of Christianity the Latin language was introduced not only in ecclesiastical but also in scientific and all important worldly transactions. Naturally the science of the Middle Ages was drawn largely from Latin sources. In fact, during the earlier of these ages Roman authors were the only ones read in the Occident. Though Greek was not wholly unknown, yet before the thirteenth century not a single Greek scientific work had been read or translated into Latin. Meagre indeed was the science which could be gotten from Roman writers, and we must wait several centuries before any substantial progress is made in mathematics.

After the time of Boethius and Cassiodorius mathematical activity in Italy died out. The first slender blossom of science among tribes that came from the North was an encyclopædia entitled *Origines*, written by **Isidorus** (died 636 as bishop of Seville). This work is modelled after the Roman encyclopædias of Martianus Capella of Carthage and of Cassiodorius. Part of it is devoted to the quadrivium, arithmetic, music, geometry, and astronomy. He gives definitions and grammatical explications of technical terms, but does not describe the modes of computation then in vogue. After Isidorus there follows a century of darkness which is at last dissipated by the appearance of **Bede the Venerable** (672–735), the most learned man of his time. He was a native of Ireland, then the home of learning in the Occident. His works contain treatises on the *Computus*, or the computation of Easter-time, and on finger-reckoning. It appears that a finger-symbolism was then widely used for calculation. The correct determination of the time of Easter was a problem which in those days greatly agitated the Church. It became desirable to have at least one monk at each monastery who could determine the day of religious festivals and could compute the calendar. Such determinations required some knowledge of arithmetic. Hence we find that the art of calculating always found some little corner in the curriculum for the education of monks.

The year in which Bede died is also the year in which **Alcuin** (735–804) was born. Alcuin was educated in Ireland, and was called to the court of Charlemagne to direct the progress of education in the great Frankish Empire. Charlemagne was a great patron of learning and of learned men. In the great sees and monasteries he founded schools in which were taught the psalms, writing, singing, computation (*computus*), and grammar. By *computus* was here meant, probably, not merely the determination of Easter-time, but the art of computation in general. Exactly what modes of reckoning were then employed we have no means of knowing. It is not likely that Alcuin was familiar with the apices of Boethius or with the Roman method of reckoning on the abacus. He belongs to that long list of scholars who dragged the theory of numbers into theology. Thus the number of beings created by God, who created all things well, is 6, because 6 is a perfect number (the sum of its divisors being $1 + 2 + 3 = 6$); 8, on the other hand, is an imperfect number ($1+2+4 < 8$); hence the second origin of mankind emanated from the number 8, which is the number of souls said to have been in Noah's ark.

There is a collection of "Problems for Quickening the Mind" (*propositiones ad acuendos iuvenes*), which are certainly as old as 1000 A.D. and possibly older. Cantor is of the opinion that they were written much earlier and by Alcuin. The following is a specimen of these "Problems": A dog chasing a rabbit, which has a start of 150 feet, jumps 9 feet every time the rabbit jumps 7. In order to determine in how many leaps the dog overtakes the rabbit, 150 is to be divided by 2. In this collection of problems, the areas of triangular and quadrangular pieces of land are found by the same formulas of approximation as those used by the Egyptians and given by Boethius in his geometry. An old problem is the "cistern-problem" (given the time in which several pipes can fill a cistern singly, to find the time in which they fill it jointly), which has been found previously in Heron, in the Greek *Anthology*, and in Hindoo works. Many of the problems show that the collection was compiled chiefly from Roman sources. The problem which, on account of its uniqueness, gives the most positive testimony regarding the Roman origin is that on the interpretation of a will in a case where twins are born. The problem is identical with the Roman, except that different ratios are chosen. Of the exercises for recreation, we mention the one of the wolf, goat, and cabbage, to be rowed across a river in a boat holding only one besides the ferry-man. Query: How must he carry them across so that the goat shall not eat the cabbage, nor the wolf the goat? The solutions of the "problems for quickening the mind" require no further knowledge than the recollection of some few formulas used in surveying, the ability to solve linear equations and to perform the four fundamental operations with integers. Extraction of roots was nowhere demanded; fractions hardly ever occur. [3]

The great empire of Charlemagne tottered and fell almost immediately after his death. War and confusion ensued. Scientific pursuits were abandoned, not to be resumed until the close of the tenth century, when under Saxon rule in Germany and Capetian in France, more peaceful times began. The thick gloom of ignorance commenced to disappear. The zeal with which the study of mathematics was now taken up by the monks is due principally to the energy and influence of one man,—**Gerbert**. He was born in Aurillac in Auvergne. After receiving a monastic education, he engaged in study, chiefly of mathematics, in Spain. On his return he taught school at Rheims for ten years and became distinguished for his profound scholarship. By King Otto I. and his successors Gerbert was held in highest esteem. He was elected bishop of Rheims, then of Ravenna, and finally was made Pope under the name of Sylvester II. by his former pupil Emperor Otho III. He died in 1003, after a life intricately involved in many political and ecclesiastical quarrels. Such was the career of the greatest mathematician of the tenth century in Europe. By his contemporaries his mathematical knowledge was considered wonderful. Many even accused him of criminal intercourse with evil spirits.

Gerbert enlarged the stock of his knowledge by procuring copies of rare books. Thus in Mantua he found the geometry of Boethius. Though this is of small scientific value, yet it is of great importance in history. It was at that time the only book from which European scholars could learn the elements of geometry. Gerbert studied it with zeal, and is generally believed himself to be the author of a geometry. H. Weissenborn denies his authorship, and claims that the book in question consists of three parts which cannot come from one and the same author. [21] This geometry contains nothing more than the one of Boethius, but the fact that occasional errors in the latter are herein corrected shows that the author had mastered the subject. "The first mathematical paper of the Middle Ages which deserves this name," says Hankel, "is a letter of Gerbert to Adalbold, bishop of Utrecht," in which is explained the reason why the area of a triangle, obtained "geometrically" by taking the product of the base by half its altitude, differs from the area calculated "arithmetically," according to the formula $\frac{1}{2}a(a+1)$, used by surveyors, where a stands for a side of an equilateral triangle. He gives the correct explanation that in the latter formula all the small squares, in which the triangle is supposed to be divided, are counted in wholly, even though parts of them project beyond it.

Gerbert made a careful study of the arithmetical works of Boethius. He himself published two works,—*Rule of Computation on the Abacus*, and *A Small Book on the Division of Numbers*. They give an insight into the methods of calculation practised in Europe before the introduction of the Hindoo numerals. Gerbert used the abacus, which was probably unknown to Alcuin. **Bernelinus**, a pupil of Gerbert, describes it as consisting of a smooth board upon which geometricians were accustomed to strew blue sand, and then to draw their diagrams. For arithmetical purposes the board was divided into 30 columns, of which 3 were reserved for fractions, while the remaining 27 were divided into groups with 3 columns in each. In every group the columns were marked respectively by the letters C (*centum*), D (*decem*), and S (*singularis*) or M (*monas*). Bernelinus gives the nine numerals used, which are the apices of Boethius, and then remarks that the Greek letters may be used in their place. [3]

By the use of these columns any number can be written without introducing a zero, and all operations in arithmetic can be performed in the same way as we execute ours without the columns, but with the symbol for zero. Indeed, the methods of adding, subtracting, and multiplying in vogue among the abacists agree substantially with those of to-day. But in a division there is very great difference. The early rules for division appear to have been framed to satisfy the following three conditions: (1) The use of the multiplication table shall be restricted as far as possible; at least, it shall never be required to multiply mentally a figure of two digits by another of one digit. (2) Subtractions shall be avoided as much as possible and replaced by additions. (3) The operation shall proceed in a purely mechanical way, without requiring trials. [7] That it should be necessary to make such conditions seems strange to us; but it must be remembered that the monks of the Middle Ages did not attend school during childhood and learn the multiplication table while the memory was fresh. Gerbert's rules for division are the oldest extant. They are so brief as to be very obscure to the uninitiated. They were probably intended simply to aid the memory by calling to mind the successive steps in the work. In later manuscripts they are stated more fully. In dividing any number by another of one digit, say 668 by 6, the divisor was first increased to 10 by adding 4. The process is exhibited in the adjoining figure. [3] As it continues, we must imagine the digits which are crossed out, to be erased and then replaced by the ones beneath. It is as follows: $600 \div 10 = 60$, but, to